

AP CALCULUS AB - UNIT 1

1.14 Connecting Infinite Limits and Vertical Asymptotes

Lesson Overview

- Interpret positive and negative infinite limits as directional unbounded behavior.
- Determine one-sided signs near denominator zeros.
- Use factor multiplicity and cancellation to distinguish holes from vertical asymptotes.
- Connect formulas, sign charts, tables, graphs, and asymptote notation.

Key Knowledge and Vocabulary

Vertical asymptote. $x = c$ is a vertical asymptote if at least one one-sided limit is $+\infty$ or $-\infty$.

Odd multiplicity. A sign-changing factor usually makes opposite branch signs.

Even multiplicity. A nonnegative small factor usually gives the same branch sign.

Cancellation first. A canceled denominator factor creates a hole, not a vertical asymptote.

Numerator sign. After simplification, combine numerator sign with every denominator factor sign on each side.

Explanation and Strategy

Step 1. Factor and cancel before building the sign chart.

Step 2. For each candidate denominator zero, choose a test point just left and just right.

Step 3. State both one-sided limits; merely listing the asymptote line is incomplete when signs are requested.

Solved Examples

Worked Example: Odd pole

Analyze $f(x) = \frac{2}{x-3}$ near $x = 3$.

Solution. The numerator is positive. The denominator is negative on the left and positive on the right, so the limits are $-\infty$ and $+\infty$.

Worked Example: Even pole

Analyze $g(x) = \frac{-5}{(x+1)^2}$ near $x = -1$.

Solution. The denominator is positive on both sides and tends to 0, while the numerator is negative. Both one-sided limits are $-\infty$.

Worked Example: Hole before pole

Analyze $h(x) = \frac{(x-2)(x+3)}{(x-2)(x-4)}$.

Solution. Cancel $x - 2$. There is a removable hole at 2 and a vertical asymptote at 4. Near 4, $x + 3 > 0$, so the left limit is $-\infty$ and the right limit is $+\infty$.

Leveled Practice

Easy Level

Foundational fluency. Focus on notation, definitions, and direct procedures.

1. Find the vertical asymptote of $1/(x - 6)$.

2. Find $\lim_{x \rightarrow 6^-} 1/(x - 6)$.

3. Find $\lim_{x \rightarrow 6^+} 1/(x - 6)$.

4. Find both one-sided limits of $1/(x + 2)^2$ at -2 .

Medium Level

Apply the idea in one or two connected steps.

5. Analyze $\frac{-3}{x+4}$ at $x = -4$.

6. Analyze $\frac{x+1}{(x-2)^2}$ at $x = 2$.

7. Analyze $\frac{x-5}{(x+1)^3}$ at $x = -1$.

8. Does $\frac{x^2-9}{x-3}$ have a vertical asymptote at 3?

Hard Level

Connect representations, conditions, and justification.

9. Find all vertical asymptotes and one-sided signs of $\frac{x+2}{(x-1)(x+3)}$.

10. Analyze $\frac{(x-2)^2}{(x-2)^3(x+1)}$ at all excluded inputs.

11. Find the one-sided limits of $\tan x$ at $x = \pi/2$.

12. For $f(x) = 1/\sqrt{x-3}$, describe the behavior at the domain boundary.

Difficult Level

Use nonroutine structure and precise mathematical reasoning.

13. For $f(x) = \frac{x-a}{(x-a)^m}$, classify the behavior at a for integer $m \geq 1$.

14. Find k so $\frac{x+k}{(x-2)(x+1)}$ has a hole rather than an asymptote at $x = 2$. Then analyze $x = -1$.

15. Analyze $\frac{1}{(x-1)^2(x+2)^3}$ at both asymptotes, including all branch signs.

Challenge Level

Synthesize, construct, generalize, or prove.

16. Construct a rational function with a hole at 1, an even pole at -2 opening downward, and an odd pole at 4 with left $+\infty$.

17. Prove that if a simplified rational function has a denominator zero of odd multiplicity and a nonzero numerator at c , then its one-sided infinite limits have opposite signs.

AP-Style Practice

Multiple-Choice Practice – No Calculator

Choose the best answer. Use exact values unless a decimal approximation is requested.

1. For $f(x) = 1/(x - 2)^2$, the one-sided limits at 2 are
(A) both $-\infty$ (B) both $+\infty$
(C) left $-\infty$, right $+\infty$ (D) finite
2. Which factor behavior produces opposite branch signs, assuming other factors stay nonzero?
(A) An even-multiplicity denominator factor (B) An odd-multiplicity denominator factor
(C) A canceled factor (D) A positive constant only
3. Before declaring a denominator zero a vertical asymptote, one should
(A) change the point value (B) factor and cancel common factors
(C) average the sides (D) ignore the numerator

Multiple-Choice Practice – Calculator Required

Choose the best answer. Use exact values unless a decimal approximation is requested.

4. A calculator graph of $\frac{x^2-4}{(x-2)^2}$ near 2 shows opposite unbounded branches. The simplified local form is
(A) $x + 2$ (B) $1/(x - 2)$
(C) $(x - 2)^2$ (D) 0
5. A table near $x = -1$ gives large negative values on both sides. Which local model is most plausible?
(A) $1/(x + 1)$ (B) $-1/(x + 1)^2$
(C) $x + 1$ (D) $|x + 1|$

Free-Response Practice – No Calculator

Let $f(x) = \frac{(x + 1)(x - 3)}{(x + 1)(x - 2)^2(x + 4)}$.

- (a) Identify the hole and its limit value. [2 points]
- (b) Identify all vertical asymptotes. [2 points]
- (c) Determine the one-sided signs at each vertical asymptote. [4 points]

Free-Response Practice – Calculator Required

A calculator table for g near $x = 1$ gives $g(0.99) = -498$, $g(0.999) = -4998$, $g(1.001) = 5002$, and $g(1.01) = 502$.

(a) Write both one-sided infinite limits. [2 points]

(b) Identify the asymptote and likely parity of the uncanceled denominator factor. [2 points]

(c) Give one simple local model consistent with the data. [2 points]